

Arpan Kumar

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EDUCATION

North Carolina State University

Ph.D. in Statistics - Third Year

Expected Graduation Date: May 2027

- GPA: 3.875/4.000
- **Advisor:** Dr. Srijan Sengupta, Associate Professor, North Carolina State University
- **Research Interests:** Network Estimation, Random Graphs, Causal Inference, Model Selection, High-Dimensional Time Series, Machine Learning, Bayesian Statistics, Financial Econometrics, and Stochastic Processes
- Teaching Assistant for **Introduction to Statistics II** in Fall 2023 and Spring 2025
- Teaching Assistant for **Statistical Methods for Researchers II** in Fall 2024 and Spring 2025
- Teaching Assistant for **Statistical Learning and Data Analytics** in Fall 2025
- Teaching Assistant for **Introduction to Data Science** in Fall 2025

Indian Statistical Institute

Master of Statistics

Class of 2021

- Applied Statistics specialization with a Finance track. Awarded Distinction with 84.85% marks
- **Subjects:** Time Series Analysis, Game Theory, Quantitative Finance, Optimization Techniques, and Statistical Machine Learning

Bachelor of Statistics

Class of 2019

- First Class Honours with 66.26% marks
- **Subjects:** Probability Theory, Linear Algebra, Estimation, Hypothesis Testing, and Statistical Inference

EXPERIENCE

Tata Digital Private Limited - Data Science Division - Bengaluru, India

Aug 2021 - Jul 2023

Data Scientist

- Developed sales forecasting models using ARIMA, SARIMA, and Prophet to account for demand shifts and structural changes following COVID-19.
- Built customer alive probability and CLTV models using BG/NBD and random forests, contributing to a 14% increase in customer loyalty and a 7% improvement in CLTV.
- Built recommendation models using Word2Vec, achieving 19% CTR and 4% CVR.
- Designed substitute and complementary recommendation algorithms to improve personalization, achieving up to 7x higher CTR, 12x higher CVR, and 28% precision / 34% recall across recommendation use cases.
- Developed A/B testing frameworks for diverse business applications using t-tests, z-tests, chi-square tests, and hypergeometric exact tests.
- Created a category-affinity metric for homepage personalization and campaign targeting, helping drive 2x higher conversions.
- Developed a popularity-based recommendation algorithm using customer transaction data and brand affinity across categories, achieving 14% CTR and 3% CVR as a fallback recommendation strategy.

General Electric - Physical Digital Analytics Lab - Bengaluru, India

May 2020 - Jun 2020

Summer Intern

- Explored methods for interpreting black-box machine learning models.
- Applied the Local Interpretable Model-agnostic Explanations (LIME) framework to practical regression and classification problems.
- Developed a more actionable approach to data segmentation using model explanation outputs.

RESEARCH PROJECTS

Ph.D. Dissertation: Predictive Subsampling for Scalable Inference in Networks

Dec 2023 - Present

- Proposed a subsample-based estimation algorithm for large and sparse RDPG networks that accurately estimates the underlying probability matrix, achieving over a 20-fold reduction in computational time with less than a 50% increase in estimation error compared to state-of-the-art methods.
- Designed a computationally efficient bootstrap-based hypothesis testing framework built on this estimator, enabling robust statistical inference while substantially reducing runtime in high-dimensional sparse networks.
- Preprint available at <https://arxiv.org/pdf/2602.16041>

Advisor: Dr. Srijan Sengupta, Associate Professor, Department of Statistics, North Carolina State University

Monotone Constrained Link Prediction for Matched Networks

Sep 2025 - Present

- Developing a method to predict missing links in a partially observed network by borrowing information from a matched fully observed network, assuming similar edge formation patterns under monotonicity constraints rather than equality of the underlying probability matrices.

Advisor: Dr. Srijan Sengupta, Associate Professor, Department of Statistics, North Carolina State University

Model Selection using Predictive Subsampling in Networks

Sep 2025 - Present

- Developing a model selection framework for different types of networks using cross-validation and predictive subsampling to reduce computational complexity.

Advisor: Dr. Srijan Sengupta, Associate Professor, Department of Statistics, North Carolina State University

Master's Thesis: Pre-birth Valuation of Racehorses Using Auction Data

Sep 2020 - Jul 2021

- Extracted data from an auction catalogue to construct a network of horses and imputed missing values.
- Developed a performance metric for racehorses using an autoencoder and an EM-like iterative algorithm incorporating pedigree data under distributional assumptions.

Advisor: Dr. Diganta Mukherjee, Professor, Sampling and Official Statistics Unit, Indian Statistical Institute

Demand Analysis with a Thin Price Sample

May 2019 - Jul 2022

- Studied the conventional sampling approach used to estimate demand elasticities of essential food items, which collects price information from many households within a first-stage unit (FSU).
- Proposed a new sampling procedure that collects price data from only one randomly selected household per FSU, thereby substantially reducing respondent burden.
- Obtained encouraging results showing that the loss in precision relative to the conventional approach was minimal despite the significant reduction in interview effort.
- Preprint available at <https://arxiv.org/abs/2206.09452>

Advisor: Dr. Diganta Mukherjee, Professor, Sampling and Official Statistics Unit, Indian Statistical Institute

OTHER ACADEMIC PROJECTS

High-dimensionality or Heterogeneity? The Bigger Problem in Federated Learning

Apr 2024

- Studied several methods, including *Avg-debias* by Lee et al., distributed learning with sparsity by Wang et al., the surrogate likelihood approach by Jordan et al., and the surrogate efficient score method by Duan et al., to understand whether heterogeneity or high dimensionality poses the greater challenge in federated learning.

Causal Inference with Clustered Data

Nov 2023

- Explored methods for analyzing causal treatment effects in the presence of cluster-level confounding in clustered data and conducted a comparative study of existing approaches.
- Investigated balance diagnostics, especially in settings with small cluster-level sample sizes.

CONFERENCES

- Received the **Statistical Programmers and Analysts Travel Grant** for presenting my work "Predictive Subsampling for Scalable Inference in Networks" at the **2026 Joint Statistical Meetings**, Boston, Massachusetts, United States.
- Attended the **2025 IMS International Conference on Statistics and Data Science (ICSDDS)**, Seville, Spain, and delivered a contributed talk on "Predictive Subsampling for Scalable Inference in Networks."
- Received the **Boyd Harshburger Travel Award** from the **Southern Regional Council on Statistics** and presented a poster on "Predictive Subsampling for Scalable Inference in Networks" at the **Summer Research Conference 2025**, Jekyll Island, Georgia, United States.

LEADERSHIP ACTIVITIES

Integration - Annual Techno-Cultural-Sports Fest of Indian Statistical Institute, Kolkata

Treasurer (2018-2020)

- Managed and maintained the accounts for the fest for two years.

Hospitality Head (2017-2018)

- Oversaw food and hospitality arrangements for guests as well as volunteers.

Volunteer Head (2016-2017)

- Managed approximately 200 volunteers during the fest.

RELEVANT SKILLS

- **Technical Skills:** Python (Pandas, TensorFlow, Keras), R, PySpark
- **Language Skills:** English (Business Professional), Hindi (Business Professional), Bengali (Native)